DATA 266 — Homework 7

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SID4 = 0670 | SEED = 670 | SLICE = 670 | HP_ID = 4 | CLS_A = 0 | CLS_B = 5

Notebook: `autoencoders_fashionmnist.ipynb`. Dataset: FashionMNIST.

Same SID/SEED as HW1. No extra HP_ID network. The two trained models are a fully connected autoencoder (FC-AE) and a VAE.

1. GANs

1.1 Architecture

A GAN has two networks. The generator G takes noise z ~ N(0, I) and outputs a fake sample. The discriminator D takes a sample and outputs the probability that it is real. They play a minimax game:

min_G max_D E[log D(x)] + E[log(1 – D(G(z)))]

Training alternates. In the D step, G is frozen and D is pushed to score real data high and G's samples low. In the G step, D is frozen and G is pushed to make D score its samples as real. In practice G maximizes log D(G(z)) instead, because early on D wins easily and the original loss gives G almost no gradient.

The goal is p_g = p_data. For a fixed G the best discriminator is D*(x) = p_data(x) / (p_data(x) + p_g(x)). Putting that back in, G is minimizing 2·JSD(p_data ‖ p_g) – log 4, which is smallest when p_g = p_data. Then D* = 1/2 everywhere and D cannot tell real from fake. Real training does not reach this. Both networks are non-convex and update at the same time, so the losses can oscillate instead of settling.

1.2 Mode collapse

Mode collapse is when G only produces a few kinds of samples even though the data has many. Many different z map to nearly the same output.

D scores each sample on its own, so nothing in the loss rewards G for being diverse. G moves to whatever D currently scores highest. D then learns to reject it, G jumps to a different point, and this repeats instead of G covering all the modes. When D is too strong or the two distributions do not overlap, the JS-based loss also gives G almost no gradient.

Two fixes:

1. **WGAN-GP.** D becomes a critic that estimates the Wasserstein distance, with no sigmoid or log. It must be 1-Lipschitz, which is enforced with a penalty $\lambda(\|\nabla D(\hat{x})\| - 1)^2$ on points between real and fake. The Wasserstein distance still gives a useful gradient when the distributions do not overlap, so the critic can be trained well without G's gradient vanishing, and G has no single sharp peak to jump to.
2. **Minibatch discrimination.** D gets features that compare each sample with the others in the same batch. A batch of near-identical samples is easy to flag as fake, which pushes G to make each batch diverse.

1.3 Three variants

DCGAN (Radford et al., 2015). The first GANs used fully connected layers and were unstable on images. DCGAN uses strided convolutions instead of pooling, transposed convolutions in G, batch norm in both, ReLU in G (Tanh at the output) and LeakyReLU in D, and no fully connected hidden layers. Training on images became stable enough to use.

Conditional GAN (Mirza & Osindero, 2014). A plain GAN gives no control over what it generates. A cGAN feeds a label y (a class, or an input image in pix2pix) to both G and D. D judges whether the sample is real for that label,

so G has to produce the class that was asked for. Pix2pix uses this with paired images and adds an L1 loss and a patch-based D.

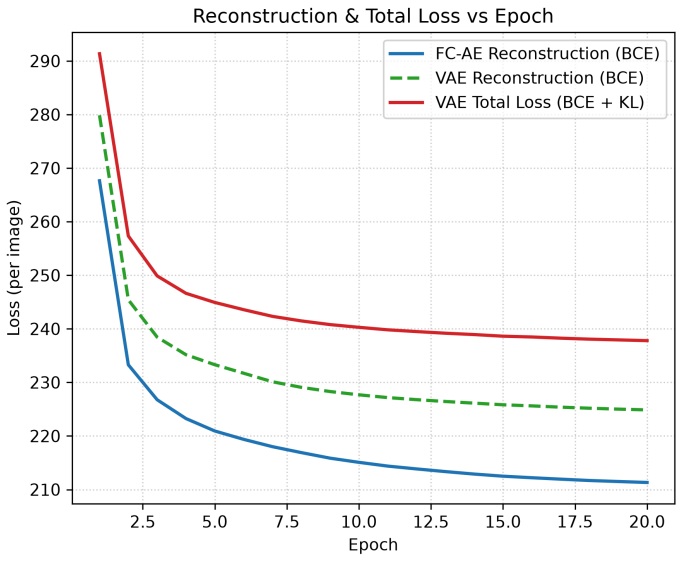
CycleGAN (Zhu et al., 2017). Image-to-image translation needed paired examples. CycleGAN trains two generators (X to Y and Y to X) and two discriminators with a cycle-consistency loss: F(G(x)) ≈ x and G(F(y)) ≈ y. That loss ties the output to the input content, so it works on unpaired sets such as horses and zebras.

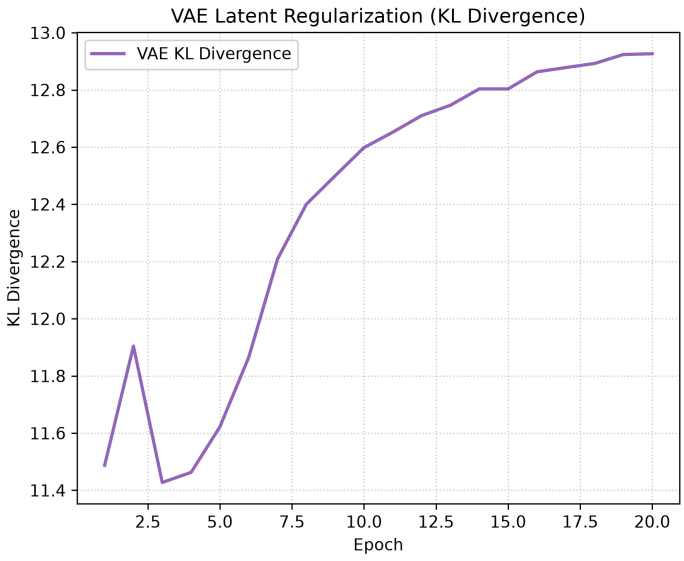
2. FC-AE vs VAE

60,000 train and 10,000 test images, 28×28, pixels 0 to 1. Both models use 784-512-128-20 for the encoder and the mirror for the decoder, sigmoid on the output. Adam, lr 1e-3, batch 128, 20 epochs, on MPS. Loss is BCE summed over the pixels and averaged per image. The VAE has μ and log σ^2 heads, samples $z = \mu + \sigma \cdot \varepsilon$, and adds the KL term to N(0, I) with weight 1.

	FC-AE	VAE
trainable params	941,092	943,672
train loss, epoch 1	267.6440	291.3547
train loss, epoch 20	211.3226	237.7973

The VAE loss is BCE + KL. At epoch 20 that is 224.8707 + 12.9265.





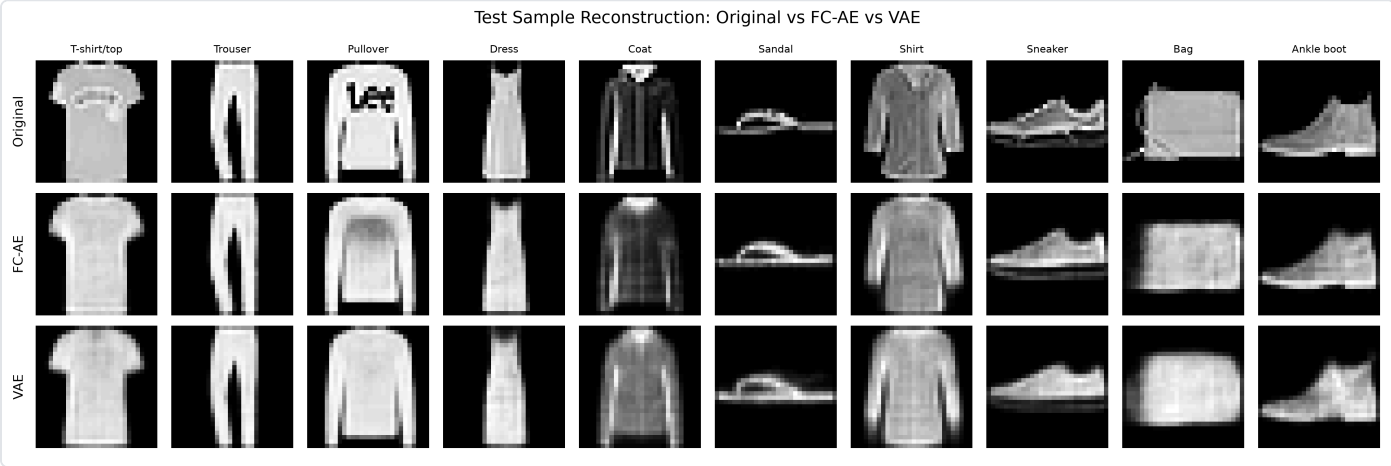
Test set

	FC-AE	VAE
Recon BCE per image	213.6070	226.8435
Recon MSE per pixel	0.009739	0.015663
KL per image	none	13.0439
BCE + KL	213.6070	239.8874

BCE + KL is not a fair comparison, since the FC-AE has no KL term.

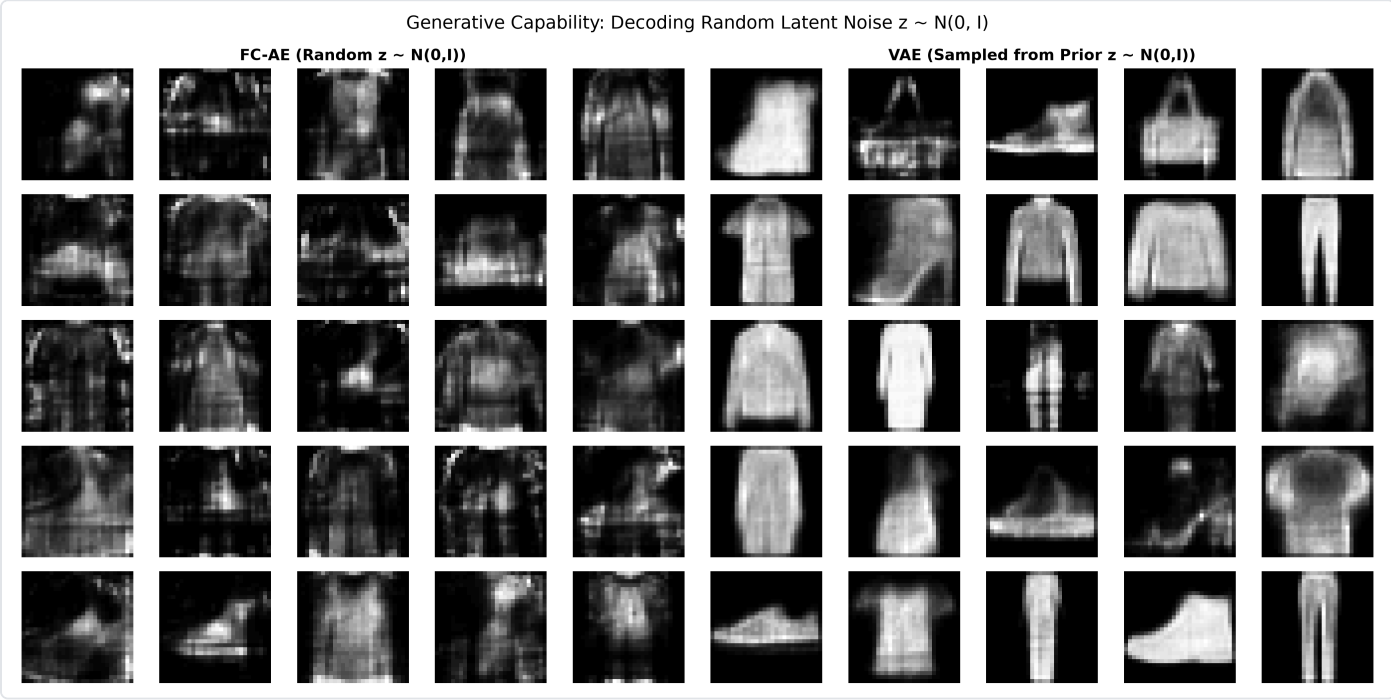
Reconstruction

FC-AE reconstructs better. Nothing pulls its code toward a prior, so all 20 dimensions go to reconstruction. In the figure the VAE copies are blurrier, especially the bag and the ankle boot. Both lose the print on the pullover. Rows are original, FC-AE, VAE.



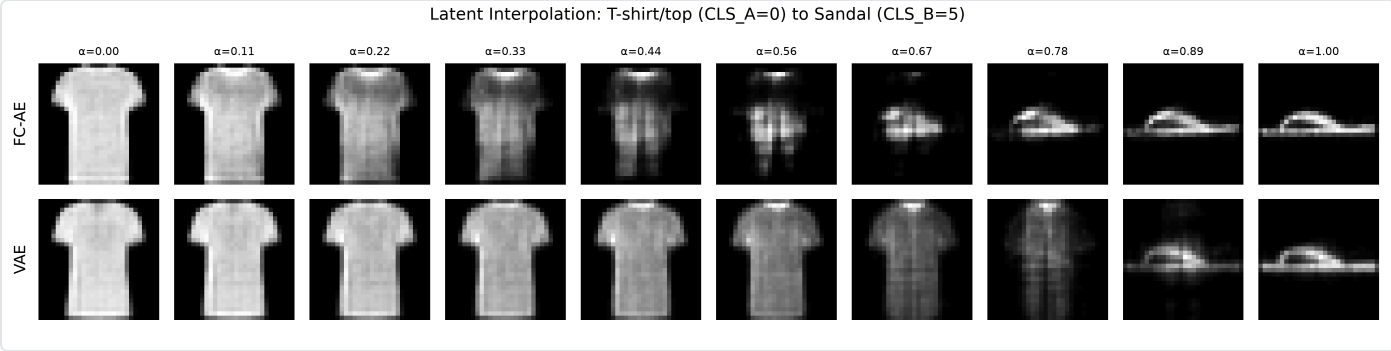
Random z

I decode z ~ N(0, I) in both. The FC-AE gives smudges and stripes, and almost none look like clothing. The VAE gives sneakers, boots, trousers, a dress, shirts and a bag, though some are blurry or mixed. The KL term pulls the VAE codes toward N(0, I), so a random z lands where the decoder was trained. The FC-AE has no reason to use that region.



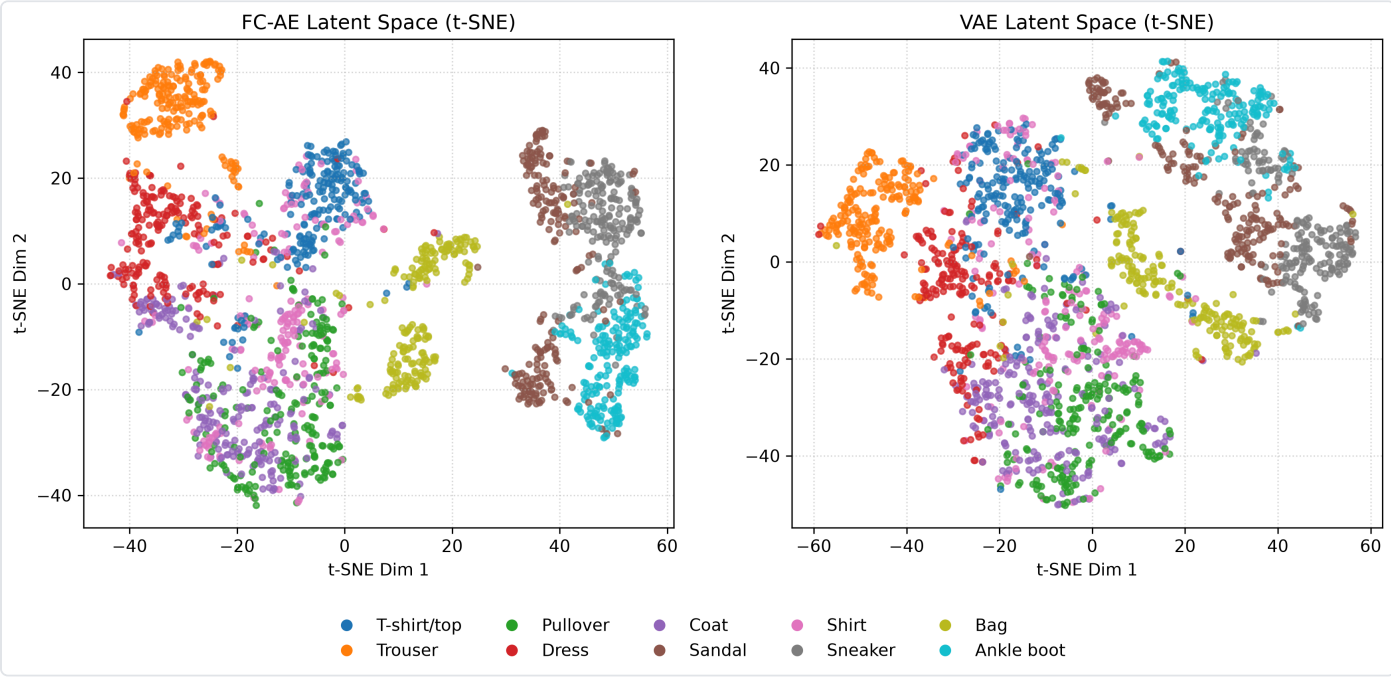
Interpolation, class 0 to class 5

First T-shirt and first sandal in the test set, 10 even steps in latent space (VAE uses μ). The top row is the FC-AE. It fades slowly: the middle frames are a smudge of shirt and sandal, and it is a sandal by about α=0.8. The bottom row is the VAE. It stays shirt-shaped and fades until about α=0.8, then is a sandal by α=0.9. I did not see a clean in-between item in either row. This is one pair, so I would not claim more.



t-SNE

2,000 test images, 20-d codes. Both plots separate trousers, bags and the shoe classes. T-shirt, pullover, coat and shirt overlap in both. I cannot see a clear difference between the two, so I am not claiming the VAE clusters are better.



Which one

For compressing and rebuilding images the FC-AE is better (MSE 0.0097 vs 0.0157). For making new images the VAE is the one that works, because its codes match the prior I sample from.